

After Action Report

Hackathon: March 13, 2025
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I. Problem

The problem tasked to us was creating a locator map for whether an address is within the PWSD Service Area that can be integrated within the PWSD website. Within the locator map if the user was within PWSD Service Area it would generate a statement, "You are within the PWSD Service Area" and if the user was not in the PWSD Service Area. "You are not within the PWSD Service".

II. Criteria and Problem Mapping

For this problem we needed to incorporate two necessary layers within the map, the Parker Water Service Area and the District Boundary. With that, we need to create a code to where the user can input an address and then find out whether they are within PWSD or not.

Pseudocode:

1. Search address (geocode into point)
2. Intersect with service area (needs to be completely within boundary)
3. Return yes (within) OR no (not within)

Steps to achieve desired result:

1. Begin by creating a basemap with the service area of PWSD
2. Write code
 - a. Geocoder (Lat & Long Coordinates)
 - b. Make sure that all projections are converted to the same one
 - i. Fix coordinate mix matches
 - c. Intersect Tool (Using polygon feature layer from AGOL)
 - i. Is the point in the polygon?

III. Tools Used

Our first route was using AGOL. We tried out ESRI Web Experience and got far in a prototype, however we got stuck at trying to create a specific output statement for when a user was within the boundary. From here we decided to move on to using python notebook within ArcPro. Looking at other resources we ultimately chose to turn to ChatGPT for help. We are still trying to learn more about Microsoft Copilot to use that tool correctly.

When using ArcGIS Experience, we utilized the "Near Me" and "Search Bar" Widgets. We were able to better learn this application, however, we realized that

this tool wasn't sufficient when trying to get two different results from one input. This led us to use ArcPro Notebook and researching which type of code would be the best way to achieve our goal. Geocoders within ArcPy would have a much more significant effect, due to its precision and its capability to use locators.

IV. Solution

While ESRI Web Experience got us on the right track it was not fully able to perform the task we wanted. We were able to learn more about ESRI Web Experience and its capabilities. In the end, creating a python code (via Chris and Theo) is the best solution. We did recommend that it could be thoughtful to add in a format for the user to see in the search box when inputting their address, an example of this would be to display, "Street Name, City Zip Code". This would eliminate confusion and potential error for the user, as well as increase accuracy in the code result.

We have included the resulting code from the various tools used and an explanation of our understanding of it on Pages 3-4. The results of the code are located on Page 5.

Overall Take-Aways and Reflection

Aiden

This Geo-Hackathon has helped me realize the importance of using the correct tools for the right project. I noticed that ESRI has more applications that could be useful such as Esri Developer. I think there are a lot of opportunities to get more familiar with what ESRI has to offer. I was able to understand how as an organization PWSD utilizes AGOL and how it's a great tool when trying to distribute geospatial information across different departments and sharing information within the organization.

Breann

Only being at the Geo-hackathon for a portion, I was able to take part in the development of the Web Experience. While I was not there for the final product, I gained more experience in AGOL. I have recently started working with AGOL so it was a great experience for me to be able to play around with more of ESRI's built-in apps and the potential capabilities they have. Along with familiarizing myself with AGOL I gained more experience in brainstorming with a team, designing a plan of action to solve a problem, and communicating with a team.

```

# Import necessary modules from the ArcGIS Python API
from arcgis.gis import GIS # For connecting to ArcGIS Online
from arcgis.geocoding import geocode # For geocoding addresses into coordinates
from arcgis.features import FeatureLayer # For accessing and querying feature layers
from arcgis.geometry import Geometry # For spatial operations like 'contains'

# --- Connect to ArcGIS Online ---
print("🔗 Connecting to ArcGIS Online...")
gis = GIS("https://www.arcgis.com", anonymous=True) # Anonymous public connection
print("✅ Connected to ArcGIS.")

# --- Get User Input ---
# Prompt the user to enter address details interactively
street_address = input('Street Address: ')
city = input('City: ')
state = input('State: ')
zip_code = input('ZIP Code: ')

# Combine inputs into a full address string
address = f'{street_address}, {city}, {state} {zip_code}'
print(f"📍 Full address entered: {address}")

# --- Geocode the Address ---
# Convert the address into geographic coordinates (latitude/longitude)
try:
    print("🔄 Attempting to geocode the address...")
    location = geocode(address)[0]['location'] # Extract coordinates of the first result
    print(f"✅ Geocoding successful: {location}")

    # Create a Geometry object from the geocoded location
    point_geom = Geometry({
        "x": location['x'],
        "y": location['y'],
        "spatialReference": {"wkid": 4326} # WGS 84 coordinate system
    })
except Exception as e:
    print(f"❌ Error: Address could not be geocoded. Details: {e}")
    point_geom = None

# --- Check if the Point is in the Polygon ---
if point_geom:
    print("🌐 Loading polygon feature layer...")
    try:
        # Connect to a hosted feature layer containing polygon boundaries
        polygon_layer = FeatureLayer(
https://services5.arcgis.com/jEt4ilNVP7EPvBYy/arcgis/rest/services/PWSD\_Service\_Area/FeatureServ
er/0
        )

```

This initial part of the code, captures each piece of the necessary layers and data for the code to properly work.

By connecting to AGOL, the code has access to the basemap and the ability to geocode.

The user input is important, in order to eliminate confusion and provide accurate results. Defining address, is beneficial to ensure the correct information is shared by the user.

Once each piece of an inputted address is collected, the code will put it together as one whole line.

This is where geocoding comes into action, the address will be created into coordinates and from there a point can be created. This also ensures that the point is projected in the same coordinate system as the basemap.

This is helpful in order to understand and troubleshoot whether an appropriate address has been entered.

These 'print' lines allow for the user to see each step of the code completed in live time. This can also be useful for troubleshooting.

This is a key point in the code, where it will be determined whether or not the point, created by geocoding, is within the service area boundary that was linked to the code through the link provided.

```
print("✅ Polygon layer loaded.")
```

```
# Query all polygons from the layer and check if any contain the point
```

```
print("🔍 Querying polygons and checking spatial containment...")
```

```
polygons = polygon_layer.query(return_geometry=True).features # Fetch features with geometry  
inside_polygon = any(  
    Geometry(poly.geometry).contains(point_geom) for poly in polygons  
)
```

```
# Output result based on spatial check
```

```
if inside_polygon:
```

```
    result = f"✅ YES: '{address}' is\033[1m WITHIN\033[0m Parker Water and Sanitation's Service  
District."
```

```
    else:
```

```
        result = f"❌ NO: '{address}' is\033[1m NOT\033[0m inside Parker Water and Sanitation's  
Service District."
```

```
except Exception as e:
```

```
    result = f"❌ Error during spatial check. Details: {e}"
```

```
else:
```

```
    result = "❌ Geocoding failed, cannot check spatial containment."
```

```
# --- Final Result Output ---
```

```
print("\n" + "="*50)
```

```
print("🏠 FINAL RESULT".center(50))
```

```
print("="*50)
```

```
print(result.center(50))
```

```
print("="*50 + "\n")
```

The print allows for the user to see that the code is checking if their address is within the service boundary.

With this query line the code will evaluate whether or not the geocoded point is within any of the polygons in the layer.

The resulting message of analysis between the geocoded point and the service boundary polygon layer.

Yes = Within Service Boundary

No = Not Within Service Boundary

Exception, this will help troubleshoot whether the point is valid or not.

Final result, will be readable and easy for the user to understand without confusion whether their address is within PWSD Boundary or not.

Moving Code Forward

The next step for this effort is to learn how to add this code to the PWSD website with a web map. The code gives the desired results, however more research might be needed. The best way of getting this interactive map on the PWSD website will need to be determined and research for that will also need to be done.

Code Results

```
🔌 Connecting to ArcGIS Online...
✅ Connected to ArcGIS.
Street Address: 13939 Ancestry Dr
City: Parker
State: CO
ZIP Code: 80134
📍 Full address entered: 13939 Ancestry Dr, Parker, CO 80134
🕒 Attempting to geocode the address...
✅ Geocoding successful: {'x': -104.825964022812, 'y': 39.50248478423}
🌐 Loading polygon feature layer...
✅ Polygon layer loaded.
🔍 Querying polygons and checking spatial containment...

=====
🔔 FINAL RESULT
=====
✅ YES: '13939 Ancestry Dr, Parker, CO 80134' is WITHIN Parker Water and Sanitation's Service District.
=====
```

```
🔌 Connecting to ArcGIS Online...
✅ Connected to ArcGIS.
Street Address: 1145 Sherman
City: Denver
State: CO
ZIP Code: 80203
📍 Full address entered: 1145 Sherman, Denver, CO 80203
🕒 Attempting to geocode the address...
✅ Geocoding successful: {'x': -104.984873759088, 'y': 39.734854902949}
🌐 Loading polygon feature layer...
✅ Polygon layer loaded.
🔍 Querying polygons and checking spatial containment...

=====
🔔 FINAL RESULT
=====
❌ NO: '1145 Sherman, Denver, CO 80203' is NOT inside Parker Water and Sanitation's Service District.
=====
```

Full address entered

Geocoded address

Geocoded status

Polygon layer visible

Polygon layer status

Checking Geocode and Polygon compatibility

Final Result

Is my address within PWSD?